

Simulations @ sultan
TFG 2025 / 2026

Connect to sultan:

ssh -X tfg_2025_pablo@sultan.unizar.es (Pwd same as user)

Work environment:

Copy /home/hgomez/.bashrc and /home/hgomez/.pc_profile to your home directory

To load these configuration files type bash in the current session or open a new one

To verify that the environment has been loaded correctly do:

which geant4-config

You should see the output: /home/hgomez/geant4.11.0.3-install/bin/geant4-config

which root-config

You should see the output: /programas/root/6.28.04/bin/root-config

G4TomoDet:

Copy /home/hgomez/G4TomoDet_TFG2025 to your working directory (it can be your home)

At your copy of G4TomoDet_TFG2025 do

```
cd build
```

```
rm CMakeCache.txt
```

```
cmake .. -DCMAKE_MODULE_PATH=$ROOTSYS/etc/cmake
```

```
make -j8
```

There will be some warnings but at the end there should be an executable G4TomoDet

Run in interactive mode (always @ build):

```
Do ./G4TomoDet
```

An OpenGL visualization window should open

At session bar type /control/execute/ ../macros/vis.mac

The geometry (a muon telescope with 4 Micromegas planes) should be displayed

Type exit to quit the visualization

To see some particles do the following sequence:

```
./G4TomoDet
```

Then at session bar

```
/control/execute/ ../macros/vis.mac  
/control/execute/ ../macros/example.mac  
/run/beamOn 10
```

After a while (~one minute) the image should be updated with some muon tracks (red lines) crossing the telescope. Once you check this, you can quit the window by typing

```
exit
```

A file G4TomoDet_TEST_example.root containing information of the 10 simulated events (simulation output) should be generated at build folder.

Change geometry and muon generator:

There are 2 files you should work on before doing simulations

G4TomoDet_TFG2025/src/DetectorConstruction.cc

Detector and Object Geometry, materials, position ...

G4TomoDet_TFG2025/src/PrimaryGeneratorAction.cc

Type, position, direction of simulated muons, ...

After any change you must compile again (make -j18)

Check geometry:

Run ./G4TomoDet ../macros/test-geo.mac > test-geo.out

At build directory test-geo.out is generated, you can check there the Y coordinates of each component (so its thickness and position in height)

Run Long Simulations:

Run ./G4TomoDet ../macros/muons.mac > muons.log

Change accordingly the muons.mac file (output filename, number of events, ...)

At build directory muons.log logfile is generated

We should decide the best way to Launch this command (screen, tmux, ...)

Option 1: Nohup

At /home/tfg_2025_pablo/G4TomoDet_TFG2025/macros tape

```
nohup ./LaunchG4.sh &
```

This command will do:

Generate at your home directory the files **date_start.dat** and **date_end.dat** (once the simulation is finished) with the timestamps of the beginning and the end of the simulation (for time control)

Launch the simulation following muons.mac (refer to the beginning of this section to adapt muons.mac to your needs). **NOTE: Do not simulate more than 105 events per simulation.**

Generate at your home directory **output-muons.log** and at /home/tfg_2025_pablo/G4TomoDet_TFG2025/macros **nohup.out** with some log information that can be useful, specially in case of problems

With this command, you can close your session and the simulation will still run.

Option 2: To be explored if Option 1 does not work

